

DEVELOPING LANGUAGE RESOURCES FOR RECOGNIZING MORAL ASPECTS IN SERBIAN LANGUAGE

MULTI-LABEL CATEGORIZATION THROUGH THE
PERSPECTIVE OF MORAL FOUNDATIONS THEORY

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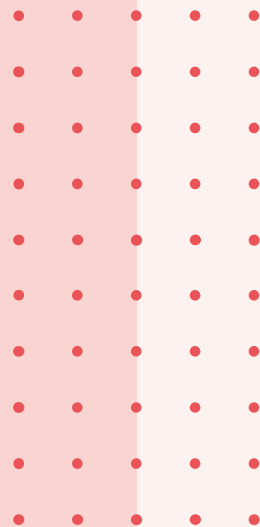
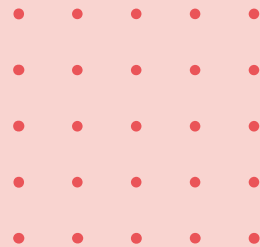


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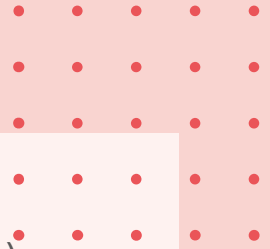
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Introduction



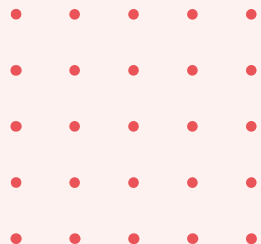
- Moral Foundations Theory (MFT) – <https://moralfoundations.org/> introduces 5 main moral foundations given with dichotomy pairs (virtue/vice):

care/harm, authority/subversion, fairness/cheating, loyalty/betrayal, purity/degradation

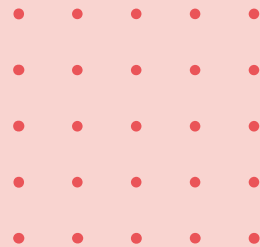
- Moral Foundations Questionnaire (MFQ) – questionnaire designed to score participant attitude toward the foundation
- Authors: Jonathan Haidt and Jesse Graham
- Theory under constant scientific realignment:
 - Other foundation candidates: liberty, honor, ownership
 - Splitting fairness into equality + proportionality (2023.)
- Used as a standard approach in computational linguistics to categorize textual sequences into moral categories

01. Objective

Develop moral language
resources for Serbian language



Study objectives



Corpus

Create corpus of conversational messages from social media platforms X and Reddit



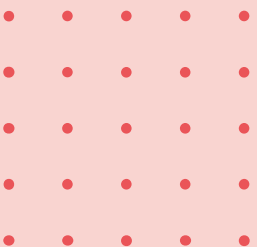
Lexicon

Identify Serbian moral words (lemmas)



AI Model

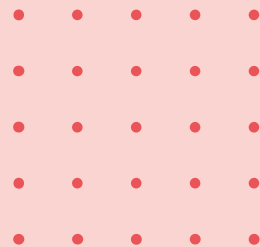
Build transformer models for moral signal recognition in Serbian texts (BERT/LLaMa)



Project Timeline

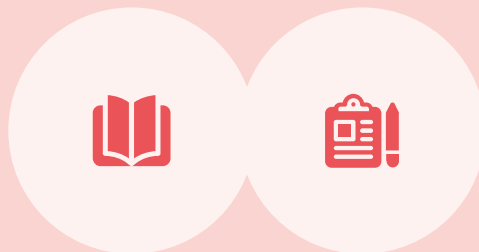


Methodology



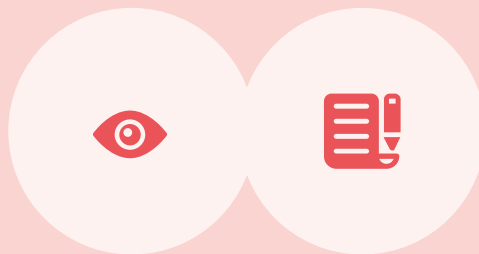
Data collection & annotation

Carefully select textual messages that contain moral signal and establish annotation schema



Annotation verification

Verify annotated sources for their validity (per annotator pairs/label/corpus)



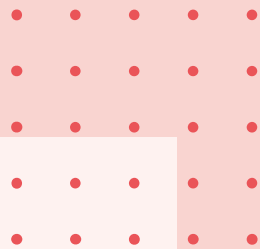
Training AI Models

Fine-tune base AI models on the classification to moral categories task

Lexicon Construction

Automatically extract moral words from the annotated corpus

Data collection & annotation



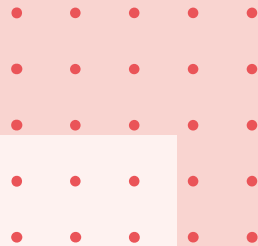
- Social-Emo.SR (~32k) → Social-Mor.SR (~13.6k)
 - Sub-corpora X-Mor.SR (~6.1k) and Reddit-Mor.SR (~7.5k)
 - Retain conversation structure (posts/comments)
 - Keywords-based filtering:

"poštenje", "pravda", "saosećanje", "zlo", "dobro", "odgovornost", "dužnost", "istina", "pravednost", "korupcija", "politika", "sloboda", "protest", "nepravda", "zdravstvo", "obrazovanje", "narod", "lider", "institucija"

(eng. "honesty", "justice", "compassion", "evil", "goodness", "responsibility", "duty", "truth", "righteousness", "corruption", "politics", "freedom", "protest", "injustice", "healthcare", "education", "people", "leader", "institutions")

- Multi-label annotation schema based on MFT categorization
- Pre-annotation with **Falcon-7b-Instruct**
- Each message was verified by 2 (different) annotators
- Established keywords-category mapping to fuel annotation consistency

Training AI Models



- Task - multi-label text classification into moral categories:
 - Moral value (foundation) – 6 categories (main foundations + non-moral)
 - Moral sentiment – 11 categories (main foundations split by sentiment + non-moral)
- **care/harm, authority/subversion, fairness/cheating, loyalty/betrayal, purity/degradation + non-moral**
- Base models – different transformer models that supports Serbian language
 - **XLM-R (base/large), Twitter-XLM-R (base/large)**
 - **Jerteh-81/355, BERTić,**
 - **LLaMa-3.2-3B-Instruct (zero-shot, fine-tuned)**
- Experiments:
 - Masking special tokens (@user, #hash, <http>)
 - Filtering out rare labelsets (< 4 assigned categories)
 - Excluding **non-moral** category (5/10 classification categories)

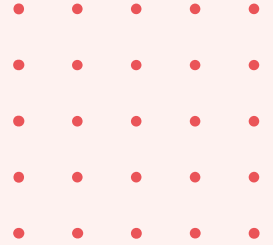
MFD.SR – Lexicon of Moral words

- Lexicon ~4.3k Serbian words
- Annotated corpus → characteristic words extraction → lexicon
- Text cleaning and preprocessing (PoS tagging, NER removal)
- SRPOL (SentiWords.SR) to determine lemmas sentiment
- Cosine similarity between the words (to identify derivations)
- c-Tf-Idf algorithm:

$$c - TF - IDF_i = \frac{t_i}{w_i} \times \log \frac{m}{\sum_j^n t_j}$$

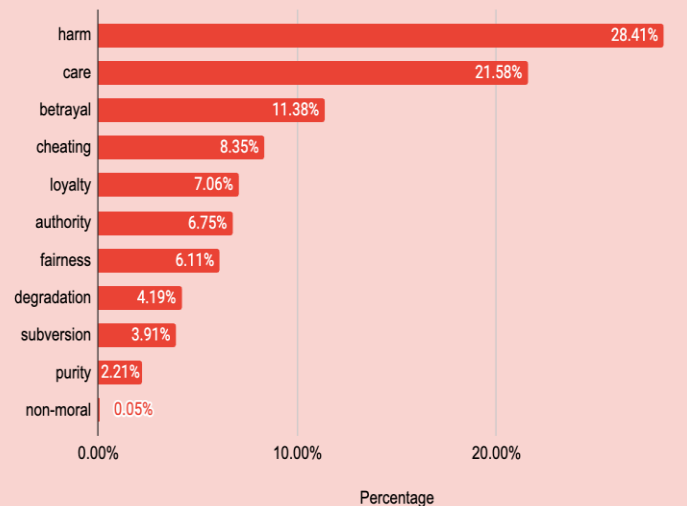
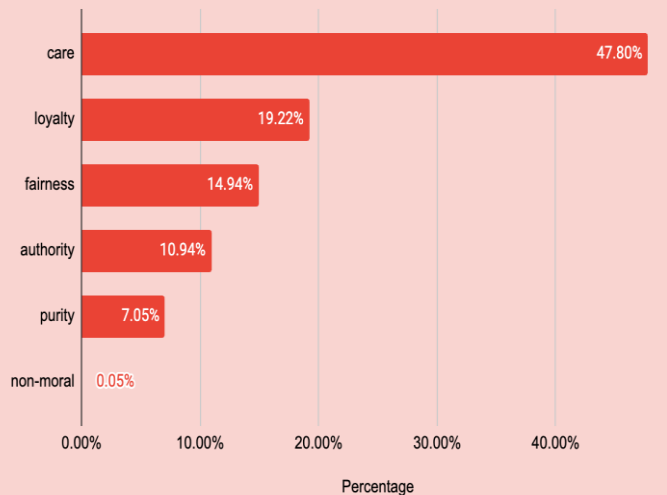
```
"MFD.SR": [  
  {  
    "lemma": diskriminacija,  
    "PoS": NOUN,  
    "weights":  
    {  
      "authority": 0.18,  
      "care": 0.16,  
      "fairness": 0.25,  
      "loyalty": 0.14,  
      "purity": 0.27  
    },  
    "sentiment": -0.64,  
    "moral_foundation": purity,  
    "moral_sentiment": purity.vice,  
    "moral_category": degradation  
  },  
  ...  
]
```

Results Evaluation



Corpus Statistics

- **Social-Mor.SR** ~13.6k messages (post/comments), ~3,3k threads
- 45.35 tokens/message, 21.76 valid tokens/message
- Labelsets: 383, Card: 1.8, $P_{\min}$: 0.52
- Average sentiment intensity (SRPOL) 0.06 (X/Reddit = 0.04/0.08)
- Care/harm the most dominant categories
- Category co-occurrence analysis: (care, harm), (cheating, harm), (betrayal, harm), (betrayal, cheating)



Annotation Verification



- Avg Cohen's kappa ~**0.23**
- Challenges in interpretation:

dichotomy pairs signals (care vs. harm), similar categories e.g. subversion/degradation/harm, in messages with metaphoric/sarcastic meaning

- Annotations harmonization with consensus labels

$$\text{Gold label} = \sum_m \text{mode}_m(A_l), \text{mode}_m(A_l | l \in L, l \neq \text{mode}_t(A), m \leq t \leq n \leq L)$$

"Zagađenje u Smederevu: Analize pokazuju povišene nivoe arsena i bora"

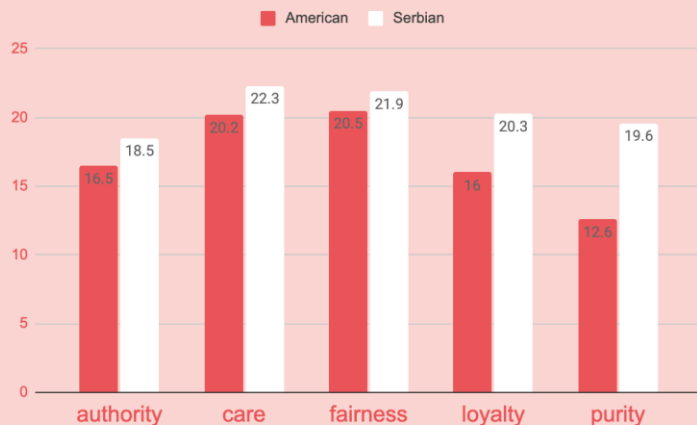
Moral sentiment: ["harm", "care", "fairness"]

AI Models trained on Social-Mor.SR

Moral-BERT.SR/Moral-LLaMa.SR

	F_1^{Ma}	F_1^{Mi}	F_1^W	Acc	HS
XLM-R-Large	0.45	0.66	0.65	0.40	0.62
Twitter-XLM-R-Large	0.47	0.68	0.66	0.41	0.63
Jerteh-355	0.42	0.67	0.64	0.39	0.61
LLaMa-3.2.-3B-Instruct - zero-shot	0.13	0.21	0.19	0.02	0.13
LLaMa-3.2.-3B-Instruct -fine-tunned	0.29	0.56	0.55	0.09	0.43

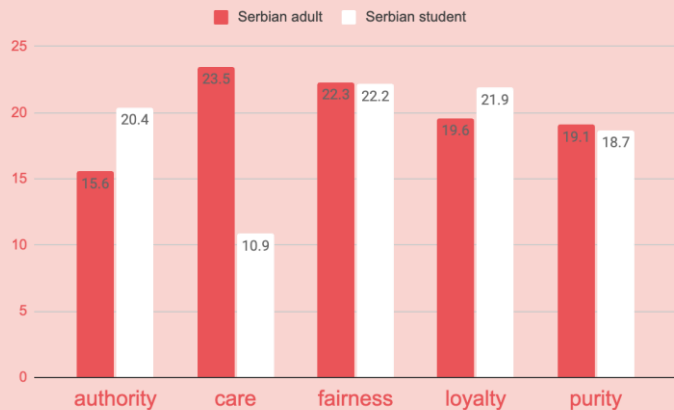
Survey on Moral Values Understanding



- Anonymous and voluntary
- 370 participants

3 groups of questions:

- Socio-demographic – 9
- Moral values understanding – 10 → test lexicon validity
- MFQ – 32 → correlate with identified moral lemmas



authority: $F_s=34.4, p \leq 0.05$
care: $F_s=138.9, p \leq 0.05$
fairness: $F_s=40.9, p \leq 0.05$
loyalty: $F_s=55.5, p \leq 0.05$
purity: $F_s=39.4, p \leq 0.05$

authority: $r=0.07, p=0.20$
care: $r=0.13, p=0.01$
fairness: $r=0.10, p=0.07$
loyalty: $r=0.11, p=0.03$
purity: $r=0.04, p=0.47$

29.7%

Volunteers from social
networks

70.3%

Volunteers among the students
from Belgrade University

Conclusion

Corpus ~13.6k

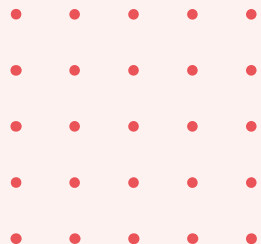
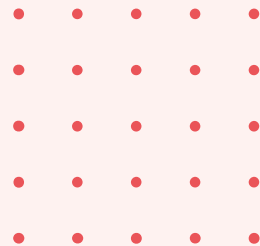
Annotated messages from social media X and Reddit

Lexicon ~4.3k

Annotated Serbian words

Models ~8

Fine-tuned transformer models (BERT/LLaMa)



Future Work

01.

Additional annotators to ensure annotations validity

AI models re-training, lexicon validation on enriched/new annotated datasets

02.



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- Bulla, L. a. (2022). Detection of morality in tweets based on the moral foundation theory, International Conference on Machine Learning, Optimization and Data Science, 1–13
- Graham, J. a. (2009). Liberals and conservatives rely on different sets of moral foundations. Journal of personality and social psychology, 96, 1029
- He, Z. a. (2024). Whose Emotions and Moral Sentiments Do Language Models Reflect?, Findings of the Association for Computational Linguistics ACL 2024, 6611–6631
- Hoover, J. a.-W. (2020). Moral foundations twitter corpus: A collection of 35k tweets annotated for moral sentiment. Social Psychological and Personality Science, 11(8), 1057–1071
- Zangari, L. a. (2025). A survey on moral foundation theory and pre-trained language models: current advances and challenges. AI & SOCIETY, 1–26

Utilized Resources

- PoS taggers/lemmatizers for Serbian language

Stanković, R., Šandrih B., Krstev C., Utvić M., and Škorić M. (2020). Machine learning and deep neural network-based lemmatization and morphosyntactic tagging for Serbian, In Proceedings of The 12th Language Resources and Evaluation Conference, pages 3954–3962

- SRPOL framework for measuring sentiment intensity in Serbian texts

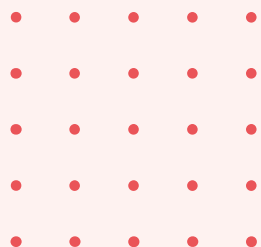
Šošić, M. (2022). SRPOL – A Lexicon Based Framework for Sentiment Strength of Serbian Texts. Review of the NCD, 41:58–73

- eMFD lexicon of moral words for English (annotated sources)

Hopp, F. et al. (2021). The extended Moral Foundations Dictionary (eMFD): Development and applications of a crowd-sourced approach to extracting moral intuitions from text, Behavior Research Methods, 53: 232–246

- REDI model for diacritics restoration

Ljubešić N., Erjavec T., and Fišer D. (2016), Corpus-based diacritic restoration for south slavic languages. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), pages 3612–3616



Utilized Resources - transformer models

- **XLM-R:**

Conneau A., Lample, G. (2019), Cross-lingual Language Model Pretraining, In 33rd Conference on Neural Information Processing Systems (NeurIPS), 32, 7059–7069

- **Twitter-XLM-R:**

Bariberi, F. a. (2022), XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond, Proceedings of the Thirteenth Language Resources and Evaluation Conference. European Language Resources Association, 258–266

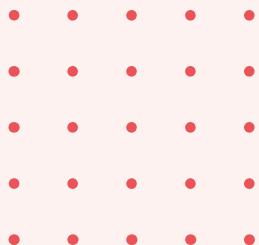
- **Jertex-81/355:**

Škorić, M. (2024). New language models for Serbian language, Infoteka, 24

- **LLaMa-3.2-3B**

Grattafiori. A. et al. (2024), The Llama 3 Herd of Models, arXiv preprint arXiv:2407.21783

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Thank you!

Do you have any questions?

Contact: milena.sosic@gmail.com

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